Waste collection routing problem with stochastic time-dependent demand increment

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Summary

- Waste Collection and Variability
- Problem Formulation
- Results
- Policies, Conclusions, and Future Works

Waste Collection and Stochastic Elements

- Overtourism





- Changing habits



- Growing (evolving) cities.

Waste Collection and Stochastic Elements



- Unespected demand (during shifts)
- Process (vehicles) disruption
- Extra shifts
- Need for new planning policies





The Problem addressed - vehicle routing problem with stochastic time-dependent demand increments

A homogeneous fleet of vehicles must visit a set of collection points with the aim of removing waste.

We divide the planning horizon (day) in not-overlapping time slots.

Each collection point must be visited exactly once during the planning horizon and it has a known initial demand which may increase along the day. The (uncertain) demand increments are revealed upon arrival at the collection points.

The goal is to minimize the expected total cost which also considers a penalty for the residual quantity of waste accumulated for the next day.

Stochastic Vehicle Routing

Routing with stochastic demand

Waste collection routing problems with stochastic demand have been extensively studied in the literature, mostly considering demand unknown but stationary.

Markov et al.[a] deal with a demand varying on time. However, they consider a multi-period setting in which demands may increase from one period to another, but it is constant over the same period (routes span over a single period, then the exact quantity is know at the beginning of the daily plan).

Conversely, we study a single day setting, where the time-horizon is split in time-slots and routes can span over multiple time-slots.

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[a] I. Markov, M. Bierlaire, J.-F. Cordeau, Y. Maknoon, S. Varone, "Waste collection inventory routing with non-stationary stochastic demands", Computers & Operations Research 113, 104798 (2020)

Problem Formulation

Let m be the number of available vehicles and G a direct graph. Nodes $1, ..., n$ of G represent the collection points, whereas node 0 and nodes $n+1, ..., n+m$ represent the central depot (same location).

Sets:

- $V = \{0, 1, ..., n, n+1, ...n+m\}$: set of all nodes
- $V' = \{0, 1, ..., n\} = V \setminus \{n+1, ..., n+m\}$
- $V'' = \{1, ..., n, n+1, ...n+m\} = V \setminus \{0\}$
- $V''' = \{n+1, ...n+m\} = V \setminus V'$
- S : set of scenarios
- T : set of periods

The Problem - Parameters

C vehicle capacity

c_{ij} travel cost associated with arc (i, j), $i \in V'$, $j \in V''$ $(j \neq i)$

t_{ij} travel time associated with arc (i, j), $i \in V'$, $j \in V''$ $(j \neq i)$.

e^t starting time of period $t \in T$

ℓ^t ending time of period $t \in T$

p unit penalty cost for uncollected waste

d_j initial non-negative (deterministic) demand $j = 1, ..., n$

δ^s_{jt} non-negative demand increment $j = 1, ..., n$, (beginning) $t \in T$ $s \in S$

M, T_{max}, Q_{max} large constant

Formulation: Variables

x_{ij} : binary variable equal to 1 if a vehicle travels from node $i \in V'$ to node $j \in V''$ $(j \neq i)$, 0 otherwise (first stage routes)

U_{jt} : binary variable equal to 1 if node $j \in V''$ is visited in period $t \in T$

$\bar{T}_j$: arrival time at node $j \in V$

y_{ij}^s : binary variable equal to 1 if a vehicle skips arc (i, j) in scenario $s \in S$, 0 otherwise

Q_j^s : theoretical cumulative load at node $j \in V$ in scenario $s \in S$

R_j^s : quantity of waste exceeding the capacity at node $j = n + 1, ..., n + m$ in scenario $s \in S$

L_j^s : quantity of waste accumulated at collection point $j = 1, ..., n$ in scenario $s \in S$ in periods following its visit, and left over for the next day

ρ_j^s : binary variable equal to 1 if node $j = 1, ..., n$ is associated with event "capacity saturation" in scenario $s \in S$, 0 otherwise

α_j^s : binary variable equal to 1 if node j is the first node associated with event "capacity saturation" in scenario $s \in S$, 0 otherwise

Example

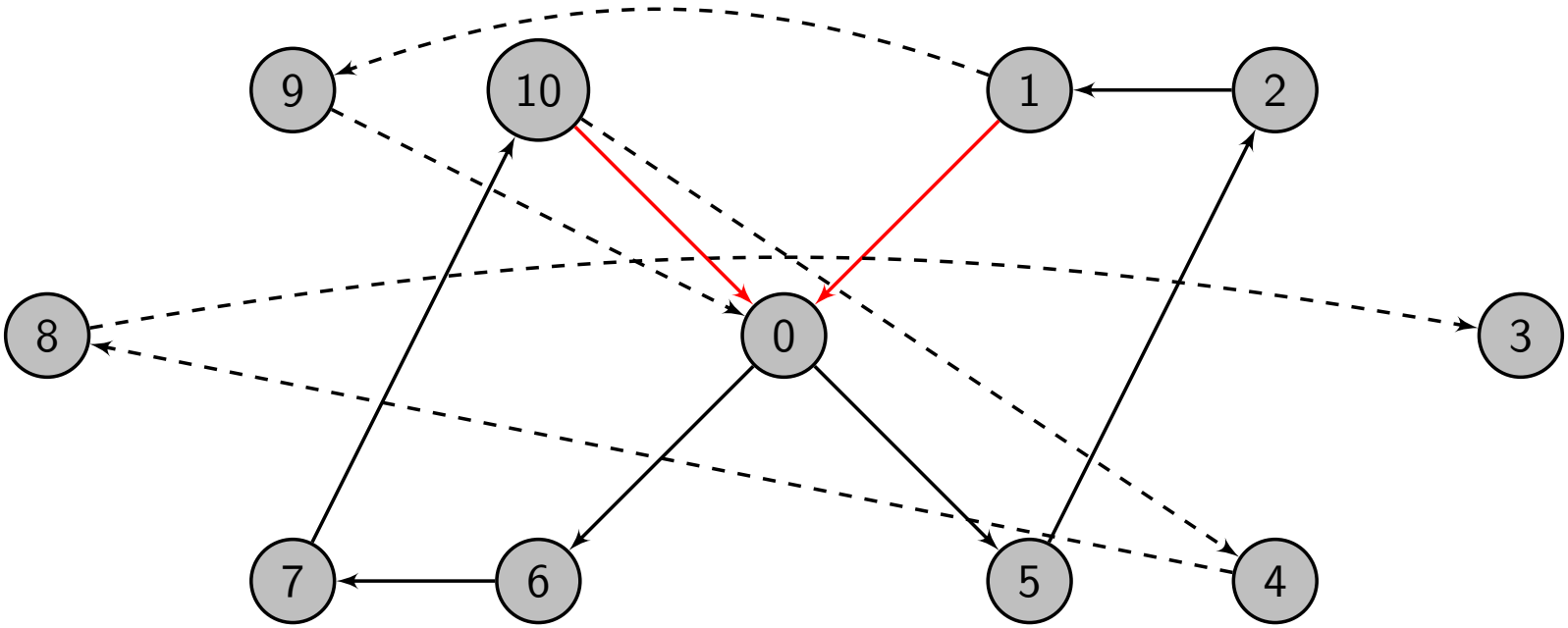


Figure: Dashed arcs ($- \rightarrow$) consider $y^s_{i,j} = 1$ and $x_{ij} = 1$, continuous arcs ($\longrightarrow$) consider $y^s_{i,j} = 0$ and $x_{ij} = 1$, while red arcs ($\longrightarrow$) are the arcs connecting the vehicles to the depot ($\alpha^s_j = 1$).

Formulation: Objective Function

$$\min \quad \sum_{i \in V'} \sum_{j \in V''(j \neq i)} c_{ij} x_{ij} - \frac{1}{|S|} \sum_{s \in S} \sum_{i=1}^{n} \sum_{j \in V''(j \neq i)} c_{ij} y_{ij}^{s} + \frac{1}{|S|} \sum_{s \in S} \sum_{j=1}^{n} c_{j0} \alpha_{j}^{s} +$$

$$+ \frac{1}{|S|} \sum_{s \in S} \sum_{j=n+1}^{n+m} pR_{j}^{s} + \frac{1}{|S|} \sum_{s \in S} \sum_{j=1}^{n} pL_{j}^{s} \qquad (1)$$

$$\min Z = Z_1 - Z_2 + Z_3 + Z_4 + Z_5$$

Formulation: Constraints (1/4)

$$\sum_{j \in V'' \setminus \{i\}} x_{ij} = \sum_{j \in V' \setminus \{i\}} x_{ji} = \sum_{t \in T} U_{it}, \quad i = 1, ..., n \tag{2}$$

$$\sum_{j \in V''} x_{0j} = m \tag{3}$$

$$\sum_{j \in V'} x_{ji} = 1, \quad i = n+1, ..., n+m \tag{4}$$

$$\sum_{t \in T} U_{jt} = 1, \quad j \in V'' \tag{5}$$

$$\bar{T}_0 = 0 \tag{6}$$

$$\bar{T}_j \geq \bar{T}_i + t_{ij} - M(1 - x_{ij}), \quad i \in V', j \in V''(j \neq i) \tag{7}$$

Formulation: Constraints (2/4)

$$e^t - T_{max}(1 - U_{jt}) \leq \bar{T}_j \leq \ell^t + T_{max}(1 - U_{jt}), \quad j \in V'', t \in T \tag{8}$$

$$Q_0^s = 0, \quad s \in S \tag{9}$$

$$Q_j^s \geq Q_i^s - M(1 - x_{ij}) + d_j + \sum_{t \in T} \sum_{\tau=1}^{t} \delta_{j\tau}^s U_{jt}, \quad s \in S, i \in V', j = 1, ..., n \tag{10}$$

$$Q_j^s \leq Q_i^s + M(1 - x_{ij}) + d_j + \sum_{t \in T} \sum_{\tau=1}^{t} \delta_{j\tau}^s U_{jt}, \quad s \in S, i \in V', j = 1, ..., n \tag{11}$$

Formulation: Constraints (3/4)

$$Q_i^s - M(1 - x_{ij}) \leq Q_j^s \leq Q_i^s + M(1 - x_{ij}), \quad s \in S, i \in V''' \tag{12}$$

$$R_j^s \geq Q_j^s - C, \quad s \in S, j = n+1, ..., n+m \tag{13}$$

$$R_j^s \geq 0, \quad s \in S, j = n+1, ..., n+m \tag{14}$$

$$L_j^s = \sum_{t \in T} \sum_{\tau=t+1}^{|T|} \delta_{j\tau}^s U_{jt}, \quad s \in S, j = 1, ..., n \tag{15}$$

Formulation: Constraints (4/4)

$$\frac{Q_j^s - C}{Q_{max}} \leq \rho_j^s \leq 1 - \frac{C - Q_j^s}{Q_{max}}, \quad s \in S, j = 1, \ldots, n \tag{16}$$

$$y_{ij}^s \leq \frac{1}{2}(x_{ij} + \rho_i^s), \quad s \in S, i = 1, ..., n, j \in V''(j \neq i) \tag{17}$$

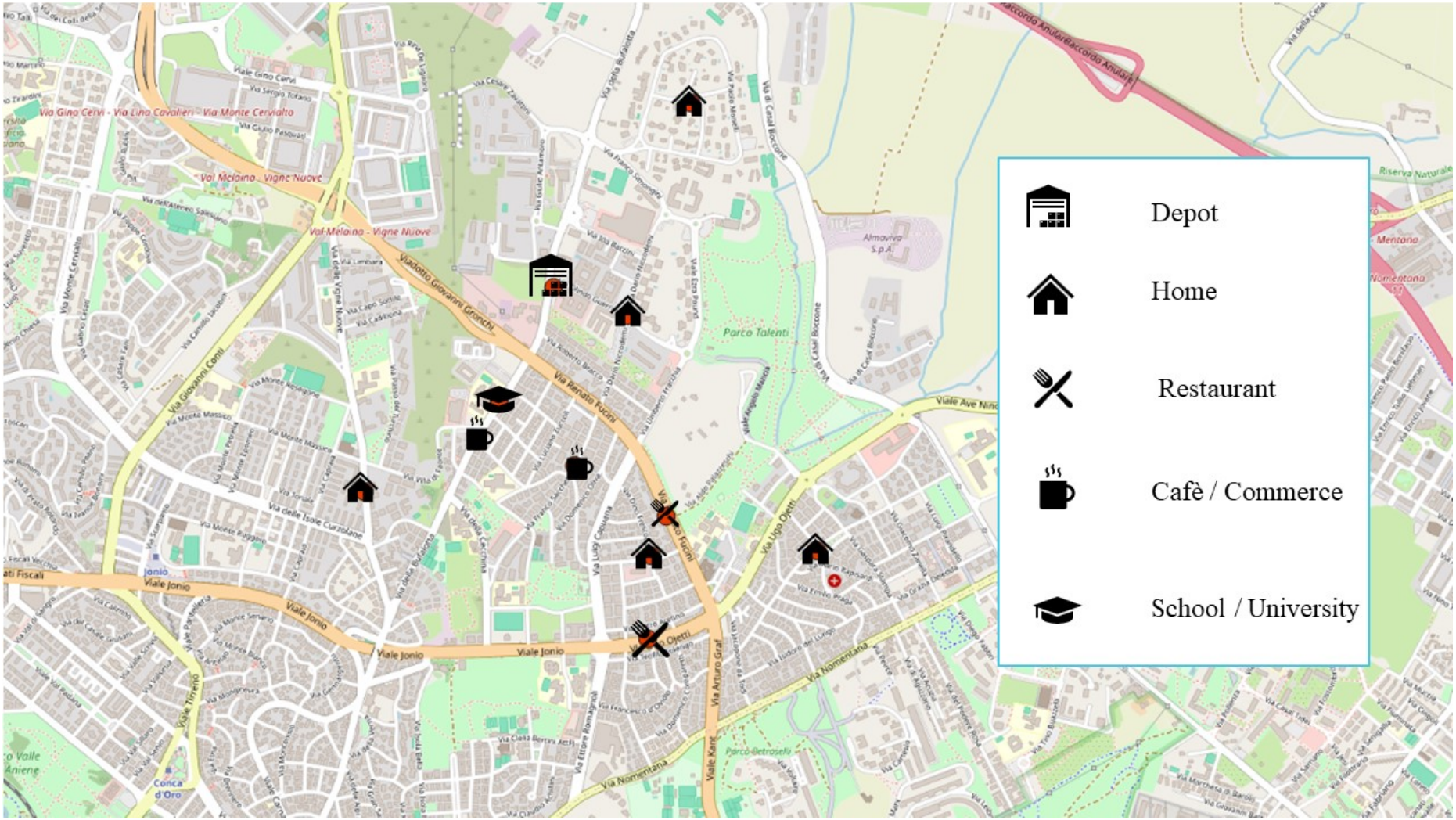
$$y_{0j}^s = 0, \quad s \in S, \quad j \in V'' \tag{18}$$

$$\alpha_j^s = \sum_{i \in V''} y_{ji}^s - \sum_{i \in V'} y_{ij}^s, \quad s \in S, \quad j = 1, \ldots, n \tag{19}$$

Realistic Instance - Different types of customers

user type	demand (d_j)	delta (δ^s_{jt}) and timeslot [e^t, l^t]			
		[8:00, 12:00]	[12:00, 16:00]	[16.00, 20:20]	[20:00, 24:00]:
home	medium	medium	low	low	medium
school	very low	low	medium	high	low
restaurant	high	very low	medium	very high	medium
bar	low	high	high	high	high

Realistic Instance



Realistic Instance: results and comments

Table: results realistic instance

Inst name	I1_R2	I1_R2_EEV
objValue	9496.31	10070.03
Z1	12607.92	11657.46
Z2	9677.65	5875.45
Z3	570.04	1444.02
Z4	3540.00	1794.00
Z5	2456.00	1050.00
Difference		6.04%

- different path / end of shift stratgy

Computational Experiments: Instances

Table: Instances parameters

Parameter	Value
n	[5,7,10]
m	3
Depot	[Central, Located lower left]
$\|T\|$	3
$\|S\|$	5
$Tmax$	[30, 50, 100]
Average D	50
Average δ	10

Patterns for customers

- *type 1*: d low; δ high, low, low
- *type 2*: d medium; δ medium, high, low
- *type 3*: d high; δ low, low, medium.

Computational Experiments: Results

Table: Results 5, 7 customers

name	objVal	EVVobjVal	Delta	runTime	gap
I2_N5_T30_C100_0	1840.29	1845.71	0.29	2.16	0.00
I2_N5_T30_C150_0	648.80	657.08	1.28	2.30	0.00
I2_N5_T30_C200_0	23.71	25.77	8.69	0.86	0.00
I2_N5_T100_C100_0	1840.29	1845.04	0.26	2.49	0.00
I2_N5_T100_C150_0	648.80	657.08	1.28	1.78	0.00
I2_N5_T100_C200_0	23.71	25.77	8.69	0.65	0.00
I2_N7_T30_C100_0	1474.07	1486.94	0.87	260.92	0.00
I2_N7_T30_C120_0	881.51	889.51	0.91	274.45	0.00
I2_N7_T30_C140_0	309.43	323.27	4.47	218.17	0.00
I2_N7_T30_C210_0	26.53	26.53	0.00	1.29	0.00
I2_N7_T100_C100_0	1474.07	1486.94	0.87	280.01	0.00
I2_N7_T100_C120_0	879.69	889.51	1.12	272.15	0.00
I2_N7_T100_C140_0	307.62	323.27	5.09	197.23	0.00
I2_N7_T100_C210_0	26.36	26.36	0.00	5.72	0.00

Computational Experiments: Results

Table: Results 5, 7 customers

name	Z1	Z2	Z3	Z4	Z5	Z1EVV	Z2EVV	Z3EVV	Z4EVV	Z5EVV
I2_N5_T30_C100_0	25.46	9.25	2.95	1664.05	157.08	22.03	7.46	3.84	1381.34	445.95
I2_N5_T30_C150_0	23.71	0.00	0.00	625.09	0.00	22.78	0.00	0.00	199.44	434.86
I2_N5_T30_C200_0	23.71	0.00	0.00	0.00	0.00	21.53	0.00	0.00	4.24	0.00
I2_N5_T100_C100_0	25.46	9.08	2.79	1821.13	0.00	22.03	8.97	4.69	1423.58	403.71
I2_N5_T100_C150_0	23.71	0.00	0.00	625.09	0.00	22.78	0.00	0.00	199.44	434.86
I2_N5_T100_C200_0	23.71	0.00	0.00	0.00	0.00	21.53	0.00	0.00	4.24	0.00
I2_N7_T30_C100_0	29.63	14.44	3.98	1394.52	60.37	35.26	19.38	2.50	1304.92	163.64
I2_N7_T30_C120_0	27.53	1.86	0.95	801.81	53.08	27.41	8.39	3.96	610.40	256.13
I2_N7_T30_C140_0	33.86	0.00	0.00	225.26	50.31	31.34	1.43	1.43	291.93	0.00
I2_N7_T30_C210_0	26.53	0.00	0.00	0.00	0.00	26.53	0.00	0.00	0.00	0.00
I2_N7_T100_C100_0	29.63	13.67	3.22	1454.89	0.00	32.18	17.88	4.08	1246.41	222.15
I2_N7_T100_C120_0	27.53	6.34	3.61	854.89	0.00	30.39	7.53	0.12	610.40	256.13
I2_N7_T100_C140_0	32.05	0.00	0.00	275.57	0.00	31.34	0.00	0.00	291.93	0.00
I2_N7_T100_C210_0	26.36	0.00	0.00	0.00	0.00	26.36	0.00	0.00	0.00	0.00

computational Experiments: results

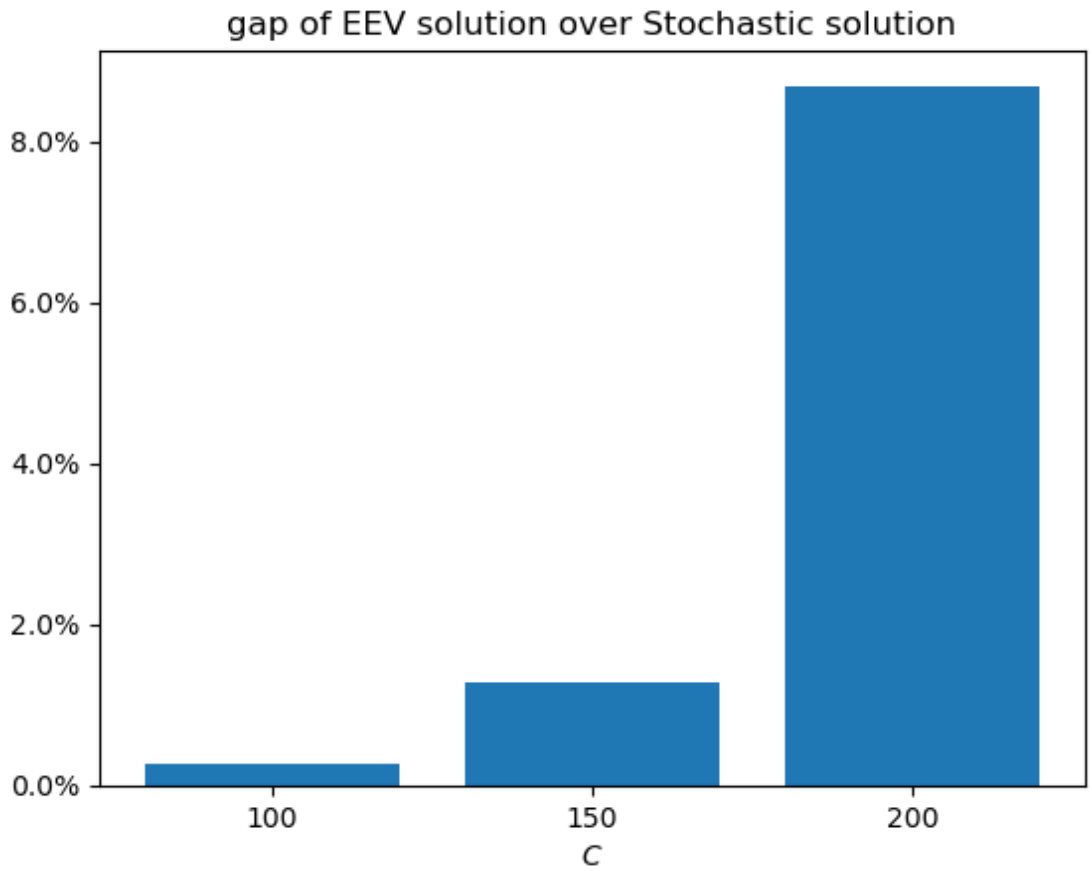


Figure: 5 customers

Computational Experiments: Results

Table: Results 10 customers

name	objVal	EVVobjVal	Delta	runTime	gap
I2_N10_T30_C275_0	91.15	140.48	54.12	555.26	0.00
I2_N10_T30_C300_0	34.73	44.94	29.39	24.55	0.00
I2_N10_T30_C325_0	33.07	33.07	0.00	15.14	0.00
I2_N10_T30_C350_0	30.17	30.20	0.10	21.38	0.00
I2_N10_T30_C400_0	30.17	30.17	0.00	7.19	0.00
I2_N10_T100_C275_0	47.70	56.63	18.73	1109.94	0.00
I2_N10_T100_C300_0	34.73	153.02	340.60	248.36	0.00
I2_N10_T100_C325_0	33.07	33.07	0.00	347.44	0.00
I2_N10_T100_C350_0	30.17	30.20	0.10	198.87	0.00
I2_N10_T100_C400_0	30.17	30.17	0.00	180.40	0.00

Computational Experiments: Results 10 customers

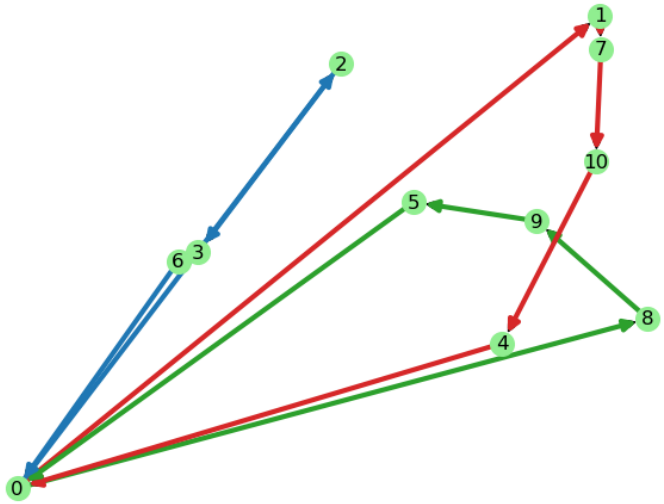
Table: Results

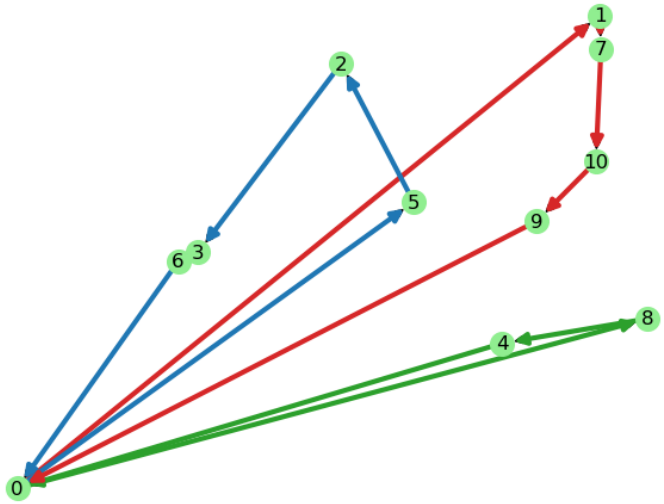
name	Z1	Z2	Z3	Z4	Z5	Z1EVV	Z2EVV	Z3EVV	Z4EVV	Z5EVV
I2_N10_T30_C275_0	40.21	0.00	0.00	9.36	41.58	42.84	1.39	1.39	97.64	0.00
I2_N10_T30_C300_0	34.73	0.00	0.00	0.00	0.00	33.22	0.00	0.00	11.72	0.00
I2_N10_T30_C325_0	33.07	0.00	0.00	0.00	0.00	33.07	0.00	0.00	0.00	0.00
I2_N10_T30_C350_0	30.17	0.00	0.00	0.00	0.00	30.17	0.00	0.00	0.03	0.00
I2_N10_T30_C400_0	30.17	0.00	0.00	0.00	0.00	30.17	0.00	0.00	0.00	0.00
I2_N10_T100_C275_0	42.65	0.00	0.00	5.05	0.00	38.17	0.00	0.00	18.46	0.00
I2_N10_T100_C300_0	**34.73**	0.00	0.00	0.00	0.00	**33.22**	0.00	0.00	0.00	**119.80**
I2_N10_T100_C325_0	33.07	0.00	0.00	0.00	0.00	33.07	0.00	0.00	0.00	0.00
I2_N10_T100_C350_0	30.17	0.00	0.00	0.00	0.00	30.17	0.00	0.00	0.03	0.00
I2_N10_T100_C400_0	30.17	0.00	0.00	0.00	0.00	30.17	0.00	0.00	0.00	0.00

Solution comments

Instance S_6

- Stochastic
  - $0 \to 1 \to 7 \to 10 \to 4 \to 12$
  - $0 \to 2 \to 3 \to 6 \to 11$
  - $0 \to 8 \to 9 \to 5 \to 13$
- EEV
  - $0 \to 1 \to 7 \to 10 \to 9 \to 11$
  - $0 \to 5 \to 2 \to 3 \to 6 \to 13$
  - $0 \to 8 \to 4 \to 12$





Policies

- **P1:** solve a standard VRP with demand equal to the initial demand (completely neglecting future demand δ_{jt}^s)
- **P2:** solve a standard VRP with demand equal to the initial demand + expected value of the sum of the additional demands delta (δ_{jt}^s). More conservative policy
- **P3:** solve a standard VRP with demand equal to the initial demand + expected value of the sum of the additional demands up to the central time slot. Less conservative than P2.

Policies

- **P4:** P2 + with vehicle capacity reduced by x% $\rightarrow$ 10% (cushion to be used for unexpected extra demand)
- **P5:** P2 + half of the vehicle have increased capacity and the other half have reduced capacity by x% $\rightarrow$ 15%
- **P6:** P2 + modified objective function (the goal is to insert the farthest nodes at the end of the routes, so in case of skipping we skip the longest arcs). min $\left[\sum_{j \in V''} \frac{1}{d_{0j}} \cdot u_j + K \sum_{j \in V} \mu_j\right]$.

Computational Experiments: Policies Results 5 customers

Table: Results policies 5 customers

name	objVal	delta	Z1	Z2	Z3	Z4	Z5
I2_N5_T30_C100_0	1840.29	0.00	25.46	9.25	2.95	1664.05	157.08
I2_N5_T30_C100_0_P1	1842.52	0.00	23.71	7.25	4.93	1196.58	624.55
I2_N5_T30_C100_0_P2	1848.66	0.00	22.03	5.85	5.19	1580.66	246.63
I2_N5_T30_C100_0_P3	1842.24	0.00	22.78	4.86	3.20	1357.84	463.29
I2_N5_T30_C100_0_P4	1878.35	0.02	21.53	5.77	5.77	1142.72	714.10
I2_N5_T30_C100_0_P5	1878.35	0.02	21.53	5.77	5.77	1142.72	714.10
I2_N5_T30_C100_0_P6	1875.17	0.02	23.99	11.35	5.72	1515.02	341.80
I2_N5_T30_C150_0	648.80	0.00	23.71	0.00	0.00	625.09	0.00
I2_N5_T30_C150_0_P1	964.38	0.49	19.89	0.00	0.00	737.82	206.67
I2_N5_T30_C150_0_P2	657.08	0.01	22.78	0.00	0.00	199.44	434.86
I2_N5_T30_C150_0_P3	650.55	0.00	25.46	0.71	0.71	350.31	274.78
I2_N5_T30_C150_0_P4	657.08	0.01	22.78	0.42	0.42	199.44	434.86
I2_N5_T30_C150_0_P5	650.55	0.00	25.46	0.00	0.00	281.94	343.15
I2_N5_T30_C150_0_P6	875.11	0.35	23.99	0.71	0.71	281.13	569.99
I2_N5_T30_C200_0	23.71	0.00	23.71	0.00	0.00	0.00	0.00
I2_N5_T30_C200_0_P1	870.08	35.70	18.40	6.68	3.78	795.00	59.58
I2_N5_T30_C200_0_P2	25.77	0.09	21.53	0.00	0.00	4.24	0.00
I2_N5_T30_C200_0_P3	60.22	1.54	19.89	0.00	0.00	40.33	0.00
I2_N5_T30_C200_0_P4	25.46	0.07	25.46	0.00	0.00	0.00	0.00
I2_N5_T30_C200_0_P5	60.22	1.54	19.89	0.00	0.00	40.33	0.00
I2_N5_T30_C200_0_P6	119.43	4.04	23.99	0.00	0.00	95.44	0.00

Computational Experiments: Policies Results 10 customers

name	objVal	delta	Z1	Z2	Z3	Z4	Z5
I2_N10_T30_C275_0	91.15	0.00	40.21	0.00	0.00	9.36	41.58
I2_N10_T30_C275_0_P1	973.50	9.68	30.17	8.43	7.97	661.31	282.47
I2_N10_T30_C275_0_P2	104.43	0.15	38.17	0.62	0.62	6.17	60.09
I2_N10_T30_C275_0_P3	976.19	9.71	32.41	4.00	4.00	432.87	510.91
I2_N10_T30_C275_0_P4	105.30	0.16	39.04	0.62	0.62	6.17	60.09
I2_N10_T30_C275_0_P5	551.11	5.05	44.86	0.00	0.00	207.15	299.10
I2_N10_T30_C275_0_P6	773.47	7.49	45.78	5.34	5.34	154.40	573.29
I2_N10_T30_C325_0	33.07	0.00	33.07	0.00	0.00	0.00	0.00
I2_N10_T30_C325_0_P1	1396.78	41.24	29.23	6.05	6.05	1153.10	214.46
I2_N10_T30_C325_0_P2	94.44	1.86	33.07	0.00	0.00	0.00	61.37
I2_N10_T30_C325_0_P3	257.05	6.77	30.17	1.64	1.64	58.79	168.09
I2_N10_T30_C325_0_P4	33.22	0.00	33.22	0.00	0.00	0.00	0.00
I2_N10_T30_C325_0_P5	181.96	4.50	32.48	0.52	0.52	149.48	0.00
I2_N10_T30_C325_0_P6	441.41	12.35	45.78	0.00	0.00	0.00	395.63
I2_N10_T30_C350_0	30.17	0.00	30.17	0.00	0.00	0.00	0.00
I2_N10_T30_C350_0_P1	918.64	29.45	29.23	5.07	5.07	720.30	169.11
I2_N10_T30_C350_0_P2	30.20	0.00	30.17	0.00	0.00	0.03	0.00
I2_N10_T30_C350_0_P3	934.62	29.98	29.23	2.58	2.58	642.71	262.68
I2_N10_T30_C350_0_P4	33.07	0.10	33.07	0.00	0.00	0.00	0.00
I2_N10_T30_C350_0_P5	239.26	6.93	30.90	1.41	1.41	32.61	175.75
I2_N10_T30_C350_0_P6	441.41	13.63	45.78	0.00	0.00	0.00	395.63

Conclusion

- waste collection / waste management important for urban grow and sustainability
- cities evolves, dynamics and uncertainty to manage in order to offer proper services
- New stochastic model for waste collection
- Difficult to solve $\rightarrowtail$ policies are easier in general.
  - lower capacity for vehicles
  - try to serve farthest users as last (fair?)
  - late collection
  - robust plan

Conclusion

Thank You!

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